

Nathan Zhang

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Education

The University of British Columbia

Sep 2023 – May 2028

B.A., Combined Major in Economics and Statistics, Minor in Data Science

Cumulative average: 93.5/100

Selected coursework: Statistical Inference (100), Machine Learning and Data Mining (93), Causal Methods (98)

Advanced Econometrics (98), Statistical Learning (97), Time Series and Forecasting (95)

Experience

ESG Research Analyst Co-op

Sep 2026 – Present

British Columbia Investment Management Corporation (BCI) / Vancouver, BC

Capital Markets & Credit Investments

- Conduct quantitative analysis to discover governance risk factors, focusing on executive compensation, firm risk, and returns. Use Databricks and Claude Code to backtest simulated governance strategies and assess shareholder interest alignment, supporting fundamental investment research, BCI's proxy voting strategy, and policy updates.

Research Assistant

May 2026 – Present

UBC Sauder School of Business | Supervisor: Allen Hu

- Evaluated financial-news coverage and missingness by matching RavenPack metadata to 3.2M+ ProQuest articles.
- Built a reproducible creator-video panel pipeline for financial-media research, with source tracking and validation.

Research Assistant

Apr 2026 – Present

UBC Sauder School of Business | Supervisors: J. Frank Li and Cheng Chen

- Independently compare measures of technological change and panel fixed-effects regression designs as sole RA on a faculty-led study, using the CPS labor survey, O*NET occupation database, and USPTO patent data.
- Use PatentsBERTa and Qwen for embeddings and feature extraction. Maintain an R/Python replication package with GMM and bootstrap inference, and write Bash and Slurm job scripts for HPC workloads.

Research Assistant

Jan 2025 – May 2026

Vancouver School of Economics at UBC | Supervisors: Jonathan Graves and Laura Nelson

- Developed Python/R teaching materials for economics and sociology courses, applying NLP, computer vision, and time-series forecasting to data science and machine learning instruction.
- Co-developed a natural language inference (NLI) tutorial with zero-shot stance classification and human-label validation, in first revision at Programming Historian. Designed and taught a computer vision workshop.

Selected Research

LLM Evaluation & Memory Efficiency

2026

Manuscript submitted, under review. Co-developed LLM memory evaluation with Jerrold Huang. In a separate 144-item benchmark, PSCC improved LLM-judged quality by about 6.9 percentage points over A-Mem while using roughly one-third fewer API tokens.

Financial NLP & Return Prediction

2026

Accepted at ICAIF 2026. Built an earnings-call panel with Jerrold Huang, linking Capital IQ transcripts, IBES, and CRSP. Tested whether management-answer novelty predicts returns beyond earnings surprise using time holdouts, placebo tests, and timing checks.

Causal Analysis of Consumer Spending

2025

Course research project. Applied cross-fitted DoubleML to about 22,000 in-person person-days. Under selection on observables and overlap, estimated a roughly 5% conditional effect of lower-friction payment mixes on daily spending.

Technical Skills

Programming & data: Python, R, SQL, Bash, Julia, PySpark, DuckDB, Polars, BQL

Methods: econometrics, factor models, causal inference, panel/time series, machine learning, NLP, computer vision

Modeling libraries: NumPy, SciPy, scikit-learn, PyTorch, statsmodels, fixest, OpenCV, spaCy, NLTK

Platforms & tools: Databricks, Bloomberg, Claude Code, DevOps/Git, GitHub Actions, HPC/Slurm, Jupyter/Quarto